The Kakeya problem and disjoint lines

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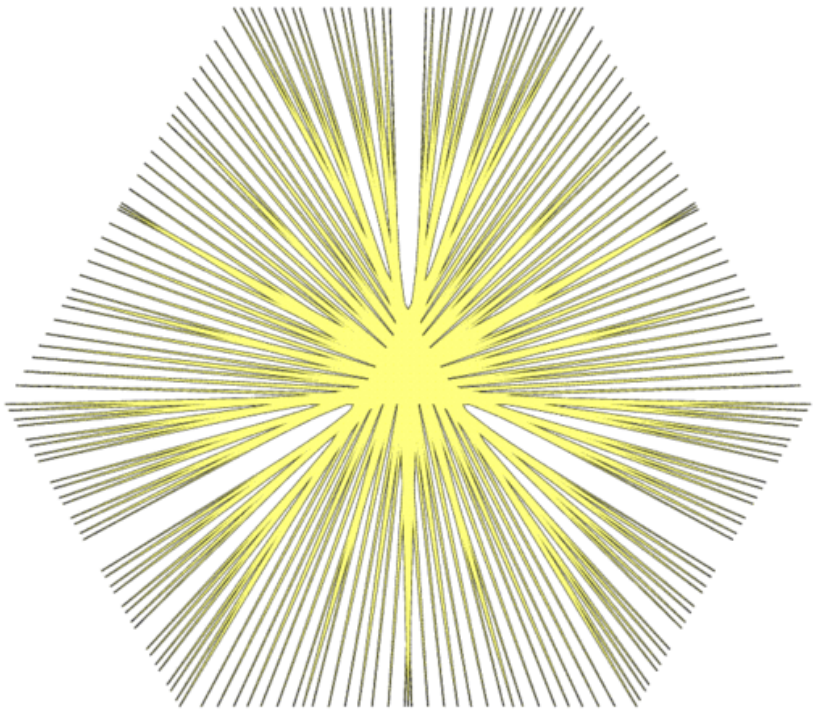
Besicovitch set

Definition

A set $B \subset \mathbb{R}^n$ is a Besicovitch set if it contains a unit line segment in every direction.

Theorem (Besicovitch 1917)

In $\mathbb{R}^n$ ($n \geq 2$) there exist Besicovitch sets of Lebesgue measure zero.





Kakeya conjecture

Definition

A set is a Besicovitch set if it contains a unit line segment in every direction.

Kakeya Conjecture

In $\mathbb{R}^n$ every Besicovitch set has Hausdorff dimension n.

Theorem (Davies 1971)

The Kakeya conjecture holds for $n = 2$.

Theorem (Zahl-Wang 2025)

The Kakeya conjecture holds for $n = 3$.

The Kakeya conjecture is open for every $n > 3$.

The current (?) best estimates for the Hausdorff dimension of a Besicovitch set $B \subset \mathbb{R}^n$ for $n \geq 4$

Definition

A set $B \subset \mathbb{R}^n$ is a Besicovitch set if it contains a unit line segment in every direction.

$n = 4$ Katz - Zahl (2019):

$$\dim_H B > 3.059$$

$n > 4$ Katz - Tao (2002):

$$\dim_H B \geq (2 - \sqrt{2})(n - 4) + 3$$

We will not improve any of these estimates.

Variants

Definition

B is a Besicovitch set if it contains a unit segment in every direction.

Kakeya Conjecture

In $\mathbb{R}^n$ every Besicovitch set B has Hausdorff dimension n.

- The Kakeya conjecture is often stated only for compact or Borel *B*.
- Often, instead of line segments, we want full lines in every direction.
- One can also require unit line segments only in a set of directions of positive $n-1$-dimensional Lebesgue measure or $n-1$-dimensional Hausdorff dimension.

It was widely belived that all of these variants are equivalent.

We proved with András Máthé that this is indeed the case: all of these forms of the Kakeya conjecture are equivalent.

Statement about the equivalence

Theorem (K-Máthé 2022)

For every n there exists a real $d \leq n$ with the following properties:

1. *There exists a compact Besicovitch set of Hausdorff dimension d.*
2. *There exists a closed set of Hausdorff dimension d that contains a line in every direction.*
3. *If $S \subset \mathbb{R}^n$ contains a line segment in a set of directions of Hausdorff-dimension $n-1$ then $\dim_H S \geq d$.*

Therefore for all possible combinations of variants we considered there exists a set of minimal dimension, and this minimal dimension is always the same for any fixed n.

Are lines much bigger than line segments?

Line Segment Extension Conjecture (K, 2015)

If A is the union of line segments in $\mathbb{R}^n$ and B is the union of the corresponding full lines, then $\dim_H A = \dim_H B$.

Theorem (K, 2015)

- *This conjecture for n would imply that for any Besicovitch set $B \subset \mathbb{R}^n$ we have $\dim_H B \geq n - 1$.*
- *If this conjecture holds for every $n \geq 2$ then any Besicovitch set has upper Minkowski dimension n.*

Theorem (K, 2015)

The Line Segment Extension Conjecture holds if $\dim_H A < 2$ or $\dim_H B \leq 2$. In particular the conjecture holds in the plane.

General Kakeya Conjecture

- Héra, K, Máthé (2018): The union of any s-dimensional family of lines in $\mathbb{R}^n$ has Hausdorff dimension $s+1$ if $s \leq 1$.

We proved with András Máthé that the following general statement is also equivalent to the Kakeya conjecture:

General Kakeya Conjecture

Let E be an aritrary subset of $\mathbb{R}^n$ and let D be the set of directions in which E contains a line segment. If $D \neq \emptyset$ then

$$\dim_H E \geq \dim_H D + 1.$$

Remarks.

- The above result implies that this conjecture holds if $\dim_H D \leq 1$.
- A possible way of improvement could be to prove it for $\dim_H D \leq t$ for some $t > 1$.
- A possible way to disprove the Kakeya Conjecture is to disprove this more general but still equivalent conjecture.

Implications between conjectures

It is not hard to show using a projective transformation that the General Kakeya Conjecture implies the Line Segment Extension Conjecture.

Thus we have the following implications:

Kakeya Conjecture $\iff$ General Kakeya Conjecture

$\Downarrow$

Line Segment Extension Conjecture

$\Downarrow$

Every Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension at least $n-1$

$\Downarrow$

Kakeya Conjecture for upper Minkowski dimension

Remarks about the implications between conjectures

Kakeya Conjecture $\Longleftrightarrow$ General Kakeya Conjecture

$\Downarrow$

Line Segment Extension Conjecture

$\Downarrow$

Every Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension at least $n-1$

$\Downarrow$

Kakeya Conjecture for upper Minkowski dimension

- For the last (green) implication we need that the assumption holds for every n. But all other implications hold for any fixed n.
- In particular, by the recent breakthrough of Zahl and Wang the Line Segment Extension Conjecture also holds in $\mathbb{R}^3$.

Typical Besicovitch sets

Theorem (T. Körner 2003)

A Besicovitch set, which is typical in the Baire category sense must have Lebesgue measure zero.
That is, in a suitable defined complete metric space the set of those Besicovitch sets that have positive Lebesgue measure is meager.

To prove the eqivalences of the several versions of the Kakeya conjecture we also used a similar approach with András Máthé although our setup was a bit different. In particular we showed the following:

Theorem (K-Máthé 2023)

In any $\mathbb{R}^n$ a typical Besicovitch set has minimal Hausdorff dimension. That is, if d is the minimal possible Hausdorff dimension of a Besicovitch set in $\mathbb{R}^n$ then in a suitable defined complete metric space the set of those Besicovitch sets that have Hausdorff dimension d is residual / co-meager.

Setup for the typical Besicovith set

With András Máthé we used the following setup. Let

$$\mathcal{K} = \{K \subset [0,1]^n \times \mathbb{RP}^{n-1} \text{ compact} : \operatorname{proj}_2 K = \mathbb{RP}^{n-1}\},$$

where proj_2 denotes projection to the second coordinate, and for any $K \in \mathcal{K}$ let

$$S(K) = \bigcup_{(p,e)\in K} \ell(p,e),$$

where $\ell(p,e)$ denotes the line through p in direction e.
The compact set K can be considered as the "code set" of a compact collection of lines in every directions, and $S(K)$ is the union of these lines.
We fix a natural metric on $\mathbb{RP}^{n-1}$ and then on the product $[0,1]^n \times \mathbb{RP}^{n-1}$ and we equip $\mathcal{K}$ with the Hausdorff metric.
We say that a property P holds for a typical Besicovitch line set if

$$\{K \in \mathcal{K} : S(K) \text{ does not have property } P\}$$

is meager.

Hausdorff dimension of a typical Besicovith line set

$$\mathcal{K} = \{K \subset [0,1]^n \times \mathbb{RP}^{n-1} \text{ compact} : \operatorname{proj}_2 K = \mathbb{RP}^{n-1}, \}$$

$$S(K) = \bigcup_{(p,e)\in K} \ell(p,e),$$

where $\ell(p,e)$ denotes the line through p in direction e.
P holds for a typical Besicovitch line set if this set is meager:

$$\{K \in K : S(K) \text{ does not have property } P\}.$$

Theorem (K-Máthé 2023)

For any n there is a real d such that in $\mathbb{R}^n$

- *d is the minimal Hausdorff dimension of Besicovitch sets (no matter which variant we consider) and*
- *a typical Besicovitch line set has Hausdorff dimension d.*

Results about typical Besicovith line sets

Theorem (K-Máthé 2023)

For any n there is a real d such that in $\mathbb{R}^n$

- *d is the minimal Hausdorff dimension of Besicovitch sets (no matter which variant we consider) and*
- *a typical Besicovitch line set has Hausdorff dimension d.*

Therefore, in order to solve the Kakeya problem, it is enough to find the Hausdorff dimension of a typical Besicovitch line set.

Now our goal is to show that if a typical Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension less than $n-1$ then it must have weird disjointness properties, which seem to be impossible.

If these properties are indeed impossible that would imply that any Besicovitch set has Hausdorff dimenson at least $n-1$.

A disjointness property of typical Besicovitch line sets

Lemma

If $B \subset \mathbb{R}^n$ is a fixed compact set such that its projection to every $n-1$-dimensional subspace has empty interior (in particular, if $\dim_H B < n-1$) then a typical Besicovitch line set in $\mathbb{R}^n$ is disjoint to B.

"Proof". We show that the (code sets of the) good collections of lines form a dense open set in $\mathcal{K}$.

By compactness, if $S(K)$ and B are disjoint then $S(K')$ and B are also disjoint whenever K' is close enough to K in the Hausdorff metric. This shows that the (code sets of the) good collections form an open set in $\mathcal{K}$.

Since the projection of B to any $n-1$-dimensional subspaces H is a compact nowhere dense set, most lines that are perpendicular to H are disjoint to B. With some extra work this gives the denseness.

More disjointness properties

Lemma

If $B \subset \mathbb{R}^n$ is a fixed compact set such that its projection to every $n-1$-dimensional subspace has empty interior (in particular, if $\dim_H B < n-1$) then a typical Besicovitch line set in $\mathbb{R}^n$ is disjoint to B.

We get immediately the following:

Corollary

If a typical Besicovitch line set in $\mathbb{R}^n$ has Hausdorff dimension less than $n-1$ then countably many typical Besicovitch line sets are pairwise disjoint.

We also get:

Corollary

If a typical Besicovitch line set in $\mathbb{R}^n$ has Hausdorff dimension less than $n-1$ then a typical Besicovitch line set consists of pairwise disjoint lines.

Combined main result

Corollary (combined)

If a typical Besicovitch line set in $\mathbb{R}^n$ has Hausdorff dimension less than $n-1$ then

- *countably many typical Besicovitch line sets are pairwise disjoint,*
- *a typical Besicovitch line set consists of pairwise disjoint lines.*

Since typical Besicovitch line sets has minimal Hausdorff dimension among Besicovitch sets and they are formed by compact collection of line sets, we obtain the following:

Theorem (Combined main result)

In $\mathbb{R}^n$ if there exists a Besicovitch set of Hausdorff dimension $< n-1$ then there exist compact collections $\mathcal{C}_1, \mathcal{C}_2, \ldots$ of line sets such that

- *each $\mathcal{C}_i$ contains lines in every directions,*
- *all the lines of $\cup_i \mathcal{C}_i$ are pairwise disjoint.*

Questions about possible constructions

Theorem (Combined main result)

In $\mathbb{R}^n$ if there exists a Besicovitch set of Hausdorff dimension $< n-1$ then there exist compact collections $\mathcal{C}_1, \mathcal{C}_2, \ldots$ of line sets such that

- *each $\mathcal{C}_i$ contains lines in every directions,*
- *all the lines of $\cup_i \mathcal{C}_i$ are pairwise disjoint.*

So the natural goal is to prove that such collections cannot exist. We do not even know if there exists one such collection.

Questions

1. *Is there a compact collection of pairwise disjoint lines in $\mathbb{R}^n$ such that we have lines in every direction?*
2. *If yes then how many such collections can one give so that their unios are pairwise disjoint?*

Weaker constructions

Questions

1. *Is there a compact collection of pairwise disjoint lines in $\mathbb{R}^n$ such that we have lines in every direction?*
2. *If yes then how many such collections can one give so that their unios are pairwise disjoint?*

What if we relax the compactness assumption?

- If we simply delete the word "compact" in the above questions then by transfinite induction it is easy to construct even infinitely many such collections.
- If we require any kind of measurability (instead of compactness) then the questions are already highly nontrivial. In this direction the best we could achieve is the following.

Theorem

There exist a closed collection of pairwise disjoint lines in $\mathbb{R}^3$ such that we have lines in every direction.

Constructions in $\mathbb{R}^3$

Theorem

There exist a closed collection of pairwise disjoint lines in $\mathbb{R}^3$ such that we have lines in every direction.

Instead of proving this theorem we first show how we can get slightly weaker results.

Using hyperboloids of one sheet we can get a closed collection of lines in $\mathbb{R}^3$ such that we get lines every non-horizontal directions.
In fact, similar constructions also work in $\mathbb{R}^n$ for any odd $n > 3$ but we do not have constructions in even dimensions.

